

LECTURE 06 · REVIEW NOTES

The Capital Asset Pricing Model

Which risk the market rewards, how total risk splits into systematic and idiosyncratic parts, and how alpha and the Security Market Line identify mispriced stocks.

About these notes

These notes review the material of Lecture 06. Each concept is explained in full, and each numerical example is worked out step by step, with every intermediate figure shown. The graded activities and interactive tools used during class are summarized in general terms and marked as in-class activities. Readings: Bodie, Kane and Marcus, Chapters 6.5, 7.1 and 7.2.

1. From the efficient frontier to the CAPM

Lectures 04 and 05 showed that combining risky assets produces an efficient frontier, and that once a risk-free asset is available every investor should hold the same risky portfolio: the tangency portfolio, the one with the highest Sharpe ratio. The Capital Asset Pricing Model (CAPM) takes that logic one step further and asks what it implies for the price, and therefore the expected return, of each individual security.

The first ingredient is diversification. Adding stocks to a portfolio lowers its volatility, but only down to a floor. The part of each stock's risk that is specific to the firm, **idiosyncratic** risk, averages out as more stocks are added. The part that comes from market-wide movements, **systematic** risk, does not: all stocks are exposed to it together. For an equally weighted portfolio of N stocks, each with volatility σ and pairwise correlation ρ , the portfolio variance is:

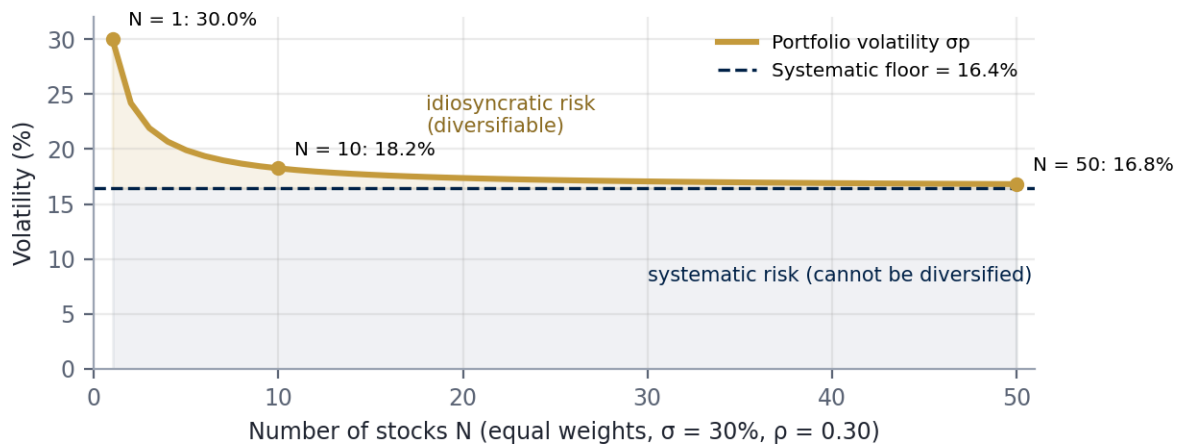
$$\sigma_p^2 = \sigma^2 \div N + (1 - 1/N) \times \rho\sigma^2 \rightarrow \text{as } N \text{ grows, } \sigma_p^2 \rightarrow \rho\sigma^2$$

WORKED EXAMPLE 1

How far can diversification go?

Each stock has $\sigma = 30\%$ and every pair has $\rho = 0.30$, so $\rho\sigma^2 = 0.30 \times 900 = 270$ (%²).

Step	Calculation	Result
N = 1	$\sigma_p^2 = 900 \div 1 + 0 \times 270 = 900$	$\sigma_p = 30.0\%$
N = 10	$\sigma_p^2 = 900 \div 10 + 0.9 \times 270 = 90 + 243 = 333$	$\sigma_p = 18.25\%$
N = 50	$\sigma_p^2 = 900 \div 50 + 0.98 \times 270 = 18 + 264.6 = 282.6$	$\sigma_p = 16.81\%$
Limit	$\sigma_p^2 \rightarrow \rho\sigma^2 = 270$	Floor = $\sqrt{270} = 16.43\%$



Portfolio volatility falls quickly with the first few stocks and then levels off at the systematic floor.

The second ingredient is equilibrium. If every investor holds the tangency portfolio, then in aggregate the tangency portfolio must contain every risky asset in proportion to its market value: it is the **market portfolio**. And because idiosyncratic risk can be eliminated at no cost by diversifying, the market does not pay investors to bear it. The only risk that earns a premium is exposure to the market portfolio, which is measured by **beta**. That conclusion is the CAPM.

IN-CLASS ACTIVITY · Diversification tool

An interactive chart in class varied the number of stocks in the portfolio and showed the portfolio volatility, the systematic floor and the share of variance that remained idiosyncratic.

2. The CAPM

$$E[r_i] = r_f + \beta_i \times (E[r_M] - r_f)$$

Component	Definition
$E[r_i]$	Expected return of security i according to the CAPM
r_f	Risk-free rate
β_i	Sensitivity of security i to the market: its systematic risk
$E[r_M]$	Expected return of the market portfolio
$E[r_M] - r_f$	Market risk premium

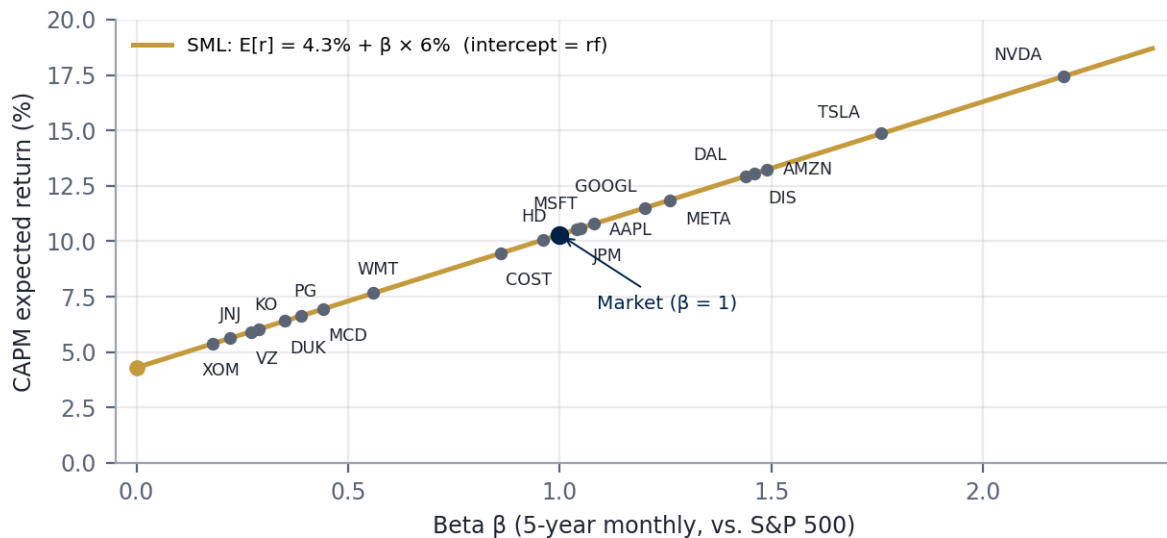
Rearranged, the model says that a security's risk premium is its beta times the market risk premium: $E[r_i] - r_f = \beta_i(E[r_M] - r_f)$. The market itself has $\beta = 1$ and earns exactly the market risk premium; a stock with $\beta = 2$ should earn twice that premium, and a stock with $\beta = 0.5$ half of it. Total volatility does not appear anywhere in the formula.

WORKED EXAMPLE 2

CAPM expected returns for real stocks

Betas were estimated from 59 monthly returns (November 2021 to September 2026) of twenty large U.S. stocks against the S&P 500, using adjusted prices from Yahoo Finance. With $r_f = 4.3\%$ and a market risk premium of 6% (close to the long-run U.S. average), the CAPM expected excess return is $\beta \times 6\%$ and the expected return is $4.3\% + \beta \times 6\%$. For NVIDIA, for example: $4.3\% + 2.19 \times 6\% = 4.3\% + 13.14\% = 17.44\%$.

Ticker	Company	Sector	β	σ	R^2	$\beta \times 6\%$	CAPM E[r]
XOM	Exxon Mobil	Energy	0.18	27.0%	1.1%	1.08%	5.38%
JNJ	Johnson & Johnson	Health care	0.22	17.8%	3.7%	1.32%	5.62%
VZ	Verizon	Telecom	0.27	22.1%	3.6%	1.62%	5.92%
KO	Coca-Cola	Consumer staples	0.29	16.1%	7.9%	1.74%	6.04%
DUK	Duke Energy	Utilities	0.35	16.2%	11.0%	2.10%	6.40%
PG	Procter & Gamble	Consumer staples	0.39	18.5%	10.5%	2.34%	6.64%
MCD	McDonald's	Restaurants	0.44	17.8%	14.5%	2.64%	6.94%
WMT	Walmart	Retail	0.56	20.6%	18.0%	3.36%	7.66%
COST	Costco	Retail	0.86	23.3%	33.2%	5.16%	9.46%
HD	Home Depot	Retail	0.96	24.1%	37.9%	5.76%	10.06%
JPM	JPMorgan Chase	Banks	1.04	24.2%	44.4%	6.24%	10.54%
MSFT	Microsoft	Software	1.05	26.5%	38.0%	6.30%	10.60%
AAPL	Apple	Technology hardware	1.08	24.9%	45.7%	6.48%	10.78%
GOOGL	Alphabet	Internet	1.20	31.3%	35.6%	7.20%	11.50%
META	Meta Platforms	Internet	1.26	44.3%	19.6%	7.56%	11.86%
DAL	Delta Air Lines	Airlines	1.44	40.2%	30.7%	8.64%	12.94%
DIS	Walt Disney	Media	1.46	33.6%	45.4%	8.76%	13.06%
AMZN	Amazon	Retail & cloud	1.49	34.2%	45.7%	8.94%	13.24%
TSLA	Tesla	Autos	1.76	58.6%	21.6%	10.56%	14.86%
NVDA	NVIDIA	Semiconductors	2.19	49.8%	46.6%	13.14%	17.44%



The twenty stocks on the Security Market Line. Under the CAPM, every expected return lies on the line and depends only on beta.

The CAPM ranking follows beta, not volatility. Exxon Mobil ($\sigma = 27.0\%$) is more volatile than Coca-Cola ($\sigma = 16.1\%$), yet its beta over this period was only 0.18, so its CAPM expected return (5.38%) is lower than Coca-Cola's (6.04%).

IN-CLASS ACTIVITY · Pick a stock

Each student selected one of the twenty stocks in an interactive Security Market Line and compared its CAPM expected excess return with the market's, adjusting the risk-free rate and the market risk premium.

3. What beta measures

Beta measures how much a stock moves, on average, when the market moves by 1%. A beta above 1 indicates a stock riskier than the market (NVIDIA 2.19, Tesla 1.76); a beta near 1, one about as risky as the market (JPMorgan 1.04, Microsoft 1.05); and a beta below 1, one less risky than the market (Coca-Cola 0.29, Johnson & Johnson 0.22).

Beta captures only **systematic** risk, for which investors are compensated. Standard deviation measures **total** risk, which includes both systematic and idiosyncratic risk; the idiosyncratic part is not compensated because it can be diversified away. This is why two stocks with the same volatility can have very different expected returns, and why a very volatile stock can still have a low expected return. Exxon Mobil is an example: $\sigma = 27.0\%$ but $\beta = 0.18$, so almost all of its volatility over the period was idiosyncratic, and its CAPM risk premium is only $0.18 \times 6\% \approx 1.1\%$.

4. Decomposing total risk

The single-index model writes a stock's excess return as a response to the market's excess return plus a firm-specific shock e_i that is uncorrelated with the market:

$$r_i - r_f = \beta_i(r_M - r_f) + e_i$$

Taking the variance of both sides, and using the fact that e_i is uncorrelated with the market, splits total variance into two parts:

$$\sigma_i^2 = \beta_i^2 \sigma_M^2 + \sigma_e^2$$

The first term, $\beta_i^2 \sigma_M^2$, is the **systematic** variance attributable to the market. The second, σ_e^2 , is the **idiosyncratic** variance, independent of market performance. The share of total variance explained by the market is a new measure of risk, R^2 :

$$R^2 = \beta_i^2 \sigma_M^2 \div \sigma_i^2 = \rho_{i,M}^2$$

R^2 lies between 0 and 1; a higher value means market risk explains most of the stock's risk. The second form follows from $\beta_i = \rho_{i,M}(\sigma_i/\sigma_M)$: substituting gives $R^2 = \rho_{i,M}^2$.

WORKED EXAMPLE 3

Decomposing the risk of two real stocks

The S&P 500 had volatility $\sigma_M = 15.5\%$ over the same period, so $\sigma_M^2 = 240.25$ ($\%^2$).

Stock	Calculation	Result
KO	Systematic = $0.29^2 \times 240.25 = 0.0841 \times 240.25$; idiosyncratic = $15.4^2 = 237.16$	20.20 + 237.16 = 257.36
	$\sigma = \sqrt{257.36}$; $R^2 = 20.20 \div 257.36$	$\sigma = 16.0\%$, $R^2 = 7.9\%$
NVDA	Systematic = $2.19^2 \times 240.25 = 4.7961 \times 240.25$; idiosyncratic = $36.4^2 = 1,324.96$	1,152.26 + 1,324.96 = 2,477.22
	$\sigma = \sqrt{2,477.22}$; $R^2 = 1,152.26 \div 2,477.22$	$\sigma = 49.8\%$, $R^2 = 46.5\%$

Even for NVIDIA, a stock with a very high beta, more than half of its variance is idiosyncratic; for Coca-Cola, over 90% is. That idiosyncratic part is what a diversified portfolio removes.

WORKED EXAMPLE 4

The Vanguard Financials ETF (VFH)

VFH has $\beta = 1.14$ and volatility 15.6%, and the S&P 500 has volatility 11.5%. How much of VFH's risk is due to market risk?

Step	Calculation	Result
1	Systematic variance = $\beta^2 \sigma_M^2 = 1.14^2 \times 0.115^2 = 1.2996 \times 0.013225$	0.017187
2	Total variance = $\sigma^2 = 0.156^2$	0.024336
3	$R^2 = 0.017187 \div 0.024336$	0.706 = 70.6%
4	$\sigma_e = \sqrt{(0.024336 - 0.017187)} = \sqrt{0.007149}$	8.46%

About 70.6% of VFH's variance comes from the market. As a diversified fund of financial stocks it has already removed much of the idiosyncratic risk of its individual holdings, so its R^2 is far higher than that of any single stock in the table above.

IN-CLASS ACTIVITY · Risk decomposition tool and Activity 1

An interactive tool in class varied β , σ_M and σ_e , with presets for real stocks, and showed how total variance splits between its systematic and idiosyncratic parts.

In Activity 1, each student received an individual case, a fictional company with its own beta and volatility together with a market volatility, a risk-free rate and an expected market return, and computed its CAPM expected return, its R^2 and its idiosyncratic volatility.

5. Alpha

To evaluate an investment strategy, investors compare its return with that of a **benchmark**: the risk-free rate, the market portfolio, or a simple buy-and-hold index strategy. The difference is the strategy's alpha. Relative to the CAPM, the benchmark is the return the model requires for the security's beta:

$$\alpha_{\text{CAPM}} = E[r] - E[r]_{\text{CAPM}}$$

A positive alpha means that, given its beta risk, the security's expected return is higher than the CAPM requires: the security is **undervalued**. An alpha of zero means it is fairly priced, and a negative alpha means it is **overvalued**.

The benchmark return is what an investor earns simply by taking market risk, a strategy that requires no skill, such as holding an index fund or ETF. Any investment strategy can therefore be decomposed as **Return = "beta" + alpha**. A passive strategy has an alpha of zero before fees: it simply earns its benchmark. An active strategy is worth pursuing, and paying for, only if it delivers a positive alpha.

6. The Security Market Line

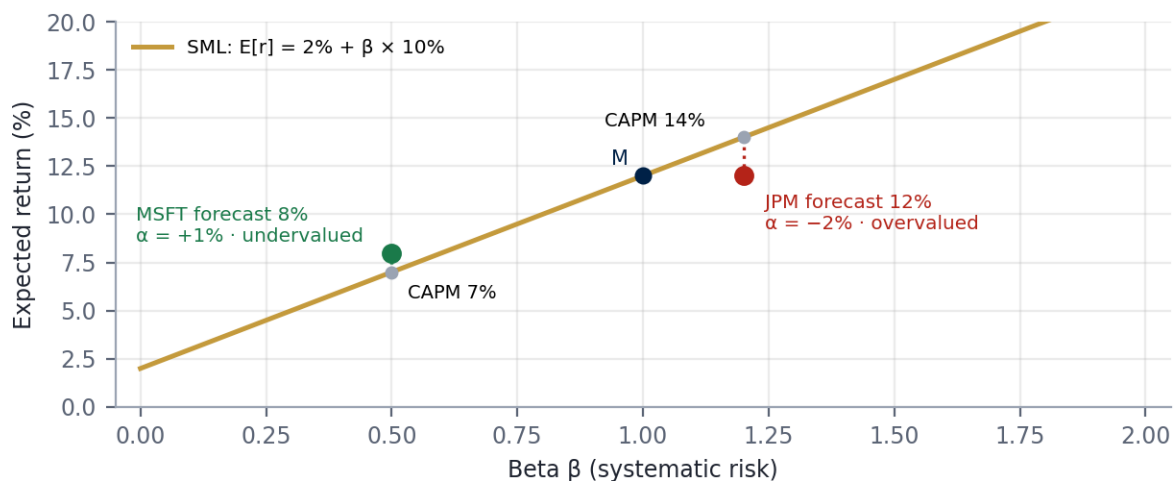
The **Security Market Line** (SML) is the graphical representation of the CAPM: expected return plotted against beta. It starts at r_f for $\beta = 0$, passes through the market portfolio at $\beta = 1$, and its slope is the market risk premium. It slopes upward because it adjusts for beta risk. Securities that plot **above** the line have positive alpha and are undervalued; those **below** have negative alpha and are overvalued; those on the line are fairly priced, whatever their beta.

WORKED EXAMPLE 5

An analyst's forecasts for JPM and MSFT

An analyst expects JPM to return 12% and MSFT 8%. The expected market return is 12%, the risk-free rate is 2%, and the betas are 1.2 for JPM and 0.5 for MSFT. Is each stock over- or underpriced?

Step	Calculation	Result
1	Market risk premium = 12% – 2%	10%
2	JPM required return = 2% + 1.2 × 10%	14%
3	JPM alpha = 12% – 14%	–2% → overvalued → sell
4	MSFT required return = 2% + 0.5 × 10%	7%
5	MSFT alpha = 8% – 7%	+1% → undervalued → buy



JPM's forecast plots below the SML (negative alpha); MSFT's plots above it (positive alpha).

JPM has the higher forecast return, yet it is the stock to sell: given its beta of 1.2, the CAPM requires 14%, and 12% falls short. MSFT's forecast is lower, but it exceeds the 7% its low beta requires.

IN-CLASS ACTIVITY · Activity 2: buy or sell?

Each student received an individual case, a fictional company with a beta and an analyst's forecast return, together with a risk-free rate and an expected market return. The task was to compute the CAPM required return and the alpha of the forecast, and to decide whether the stock was undervalued or overvalued.

7. Portfolio beta

The beta of a portfolio is the weighted average of the betas of its holdings, and the beta of cash (the risk-free asset) is zero:

$$\beta_p = w_1\beta_1 + w_2\beta_2 + \dots + w_N\beta_N$$

WORKED EXAMPLE 6

Portfolio and zero-beta positions with real betas

Step	Calculation	Result
1	50% NVIDIA + 50% Coca-Cola: $\beta_p = 0.5 \times 2.19 + 0.5 \times 0.29 = 1.095 + 0.145$	$\beta_p = 1.24$
2	\$100 long NVIDIA has dollar beta 100×2.19	\$219
3	Short \$219 of an S&P 500 ETF ($\beta = 1$): dollar beta $219 \times 1.00 = \$219$; net = $219 - 219$	Dollar beta = 0

Because betas add up, a portfolio can be built with **zero market risk**, a zero-beta or market-neutral portfolio whose value does not respond to market movements. This is the logic behind the long/short funds of Lecture 05: they hedge the market's direction and keep the stock-selection bet. What remains is idiosyncratic risk and, if the stock picks are good, alpha.

8. Key takeaways

Only beta is priced. Diversification removes idiosyncratic risk down to a systematic floor. In equilibrium everyone holds the market portfolio, so the CAPM rewards only exposure to it: $E[r_i] = r_f + \beta_i(E[r_M] - r_f)$.

Beta and volatility measure different things. Volatility is total risk; beta is only the systematic part. A highly volatile stock can have a low beta and a low expected return.

Total risk decomposes. $\sigma_i^2 = \beta_i^2\sigma_M^2 + \sigma_e^2$, and $R^2 = \beta_i^2\sigma_M^2/\sigma_i^2 = \rho^2$ is the share of variance explained by the market.

Alpha identifies mispricing. $\alpha = E[r] - E[r]_{\text{CAPM}}$. On the Security Market Line, points above the line are undervalued (buy) and points below are overvalued (sell).

Betas add up. Portfolio beta is the weighted average of the holdings' betas, which makes it possible to build market-neutral portfolios.

Betas, volatilities and R^2 for the twenty stocks were estimated for Lecture 06 from monthly adjusted prices (Yahoo Finance), November 2021 to September 2026, against the S&P 500. The VFH and JPM/MSFT figures are those presented in the lecture. Educational examples, not recommendations.